



A novel interval-based hybrid framework for crude oil price forecasting and trading

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ABSTRACT

Existing research has demonstrated the effectiveness of hybrid models in improving the accuracy of crude oil forecasting compared to single models. However, these works usually focus on point-valued crude oil closing prices which may suffer from information loss. Instead, this paper proposes a novel interval-based framework based on the principle of “divide and conquer”. After deploying variational mode decomposition (VMD) on an original training series to decompose it into low- and high-frequency components, a newly proposed autoregressive conditional interval (ACI) model is applied to predict the interval-valued low-frequency component which is treated as an inseparable random set, while the interval-valued high-frequency component is predicted by interval long short-term memory (iLSTM) networks. Combination of the two parts yields the final interval-valued prediction. A trading strategy for interval-valued data is designed and executed on a daily basis. Compared to benchmark models and competing trading strategies, the proposed framework can generate superior forecasts and deliver enhanced trading performances. The analysis within this study indicates that the framework's outstanding performance is robust to various forecasting horizons.

1. Introduction

Given that the cost and availability of crude oil exert a substantial impact on many nations' economy, political situation, and military strength, tracking and forecasting information about crude oil prices is a significant part of countries' socioeconomic and social security plans (He et al., 2010b; Yu and Ma, 2021). The forecasting of crude oil price movement has aroused great attention around the world (Huang et al., 2009). For governments, accurate predictions of crude oil price movements are instrumental in making informed energy resource reserve decisions; for crude oil-related enterprises, especially air cargo transport and petroleum-based manufacturing companies, marketing plans are heavily influenced by the price dynamics of crude oil; investors seeking to maximize profits also value such prediction accuracy (He et al., 2012; Deng et al., 2020).

The price of crude oil is subject to variation because of unforeseen natural disasters, sudden shifts in government policies, and demand and supply fluctuations, among others (Manickavasagam et al., 2020). The complex volatility of international crude oil prices, evident in any historical retrospective, include factors of nonlinearity, uncertainty

and dynamics. In addition, bubbles may emerge in crude oil price series (Phillips and Yu, 2011). Moreover, the COVID-19 pandemic has introduced new challenges to crude oil price forecasting (Narayan, 2022). Several scholars have conducted research on crude oil price forecasting during the COVID-19 pandemic. Notably, Niu et al. (2022) found that uncertainty measures are a valuable and rich information source for predicting the oil market volatility; Tian et al. (2023) illustrated that during the COVID period, combination methods furnish more accurate information than shrinkage methods. The reason is that the outbreak of the epidemic has altered the correlation between specific predictors and crude oil prices, and shrinkage methods are incapable of identifying this change, resulting in the loss of information. These contribute to making crude oil price forecasting extremely complicated and the results highly uncertain, which could cause uncertainty for the returns of related investors and for steady socioeconomic development (Zhang et al., 2015).

Although most studies focus on modeling point-valued crude oil closing prices, the method of using classic point-valued closing prices in a given trading period to model asset price dynamics can suffer

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from loss of volatility information (Yang et al., 2016). To mitigate this problem, much recent literature attempts to model the price process with an interval time series (ITS) sample and associated tools for modeling interval data (Billard, 2007; Neto and de Carvalho, 2010; Yang and Han, 2015; Dias and Brito, 2017). The ITS sample holds an advantage over a point-valued time series as it contains both the trend or level information and volatility information. Caginalp and Laurent (1998) explored the dynamics of the S&P 500 by analyzing daily data, which encompassed opening, closing, high, and low prices. Their findings illustrated that traders, influenced by price fluctuations, utilize them as indicators of their counterparts' positions, especially in relation to the equilibrium of supply and demand. In addition, the trend–volatility relationship contains valuable information in modeling the price process (Yang et al., 2016).

Within interval-valued forecasting, most efforts of late combine the “divide and conquer” concept with machine learning models, creating hybrid models that both capture internal factors and improve forecasting accuracy (Yang et al., 2022). Whereas hybrid models are less commonly applied to interval-valued crude oil price predictions, they are inherently suitable for this domain. To the best of our knowledge, only two studies rely on hybrid models for forecasting interval-valued crude oil prices (Sun et al., 2018, 2022). When (Sun et al., 2018) first proposed an interval decomposition ensemble (IDE) learning approach in this regard, their study integrated bivariate empirical mode decomposition (BEMD), interval multilayer perceptron (MLP) and interval exponential smoothing method. Sun et al. (2022) subsequently followed up by employing three-way clustering and decomposition ensemble learning to produce the interval crude oil price predictions. In the modeling process, both studies treated interval data as two points. Consequently, internal information contained in the interval data was insufficiently used and the results also lacked depth regarding trading strategies.

Interval data, composed of the lowest and highest prices during a time period, has been used in some earlier studies (He et al., 2010a; Sun et al., 2018; He et al., 2021; Sun et al., 2022). Although prediction accuracy was improved, the result was unhelpful for those investors seeking to make profits based on the predictions, given that there was still uncertainty whether the crude oil price can reach the predicted upper or lower bound. In this paper, by contrast, we construct an interval-valued price series with as a left bound the opening price and a right bound the closing price of a trading day for three important reasons:

(a) Given the deviation of the opening prices from the prior day's closing prices, integrating the information from the opening and closing prices can augment the precision of volatility estimates beyond the conventional close-to-close-based forecasting. By integrating the interval data formed by the daily opening and closing prices, we can simultaneously leverage both the overnight return (close-to-open) and the daytime return (open-to-close) for predicting both prices and volatilities. There exists a relatively large body of evidence to confirm that incorporating both opening and closing price information for volatility prediction is more effective than estimating based solely on closing prices (Garman and Klass, 1980; Gallo and Pacini, 2000; Modisett and Maboudou-Tchao, 2010). For example, Garman and Klass (1980) demonstrated that, compared to the conventional close-to-close volatility, including the opening price in the estimation procedure can reduce the variance of volatility estimates by half under specific assumptions. They also pointed out that even when using high–low prices for volatility prediction can achieve lower variance estimates, additionally including opening and closing information in the estimation can further reduce the estimation error by approximately 20%. Moreover, Fuertes et al. (2015) found that in GARCH models, utilizing the realized power variation (RPV) composed of opening and closing prices can significantly enhance the forecast accuracy of stock volatility. These studies provide compelling evidence for the effectiveness of

simultaneously using both opening and closing prices in capturing price fluctuations compared to relying solely on closing prices.

(b) The opening price is crucial for price forecasting. For example, Ibikunle (2015) pointed out that for stocks with high trading volume, the opening price provides essential additional information. Therefore, including the opening price in forecasting models can significantly enhance prediction accuracy. They also demonstrated that opening prices may affect trader sentiment throughout the day which in turn influences the day's prices. Fiszeder et al. (2023) also highlighted that the dynamics of the opening jump (the difference between today's opening price and yesterday's closing price) is arguably different from the dynamics of the trading part of the day. Thus, the opening price plays a vital role in forecasting crude oil prices.

(c) Since the magnitudes of opening and closing prices have directional attributes, the interval data formed by the opening and closing prices can provide directional information to investors, which is not achievable with interval data formed by the highest and lowest prices. Using S&P 500 daily data, Caginalp and Laurent (1998) demonstrated that traders are influenced by price movements and use them as an indication of other traders' positions, especially in terms of the balance between supply and demand. Similarly, Fiess and MacDonald (2002) pointed out that within Candlestick analysis (a popular form of technical analysis that combines Open, High, Low, and Close prices) and other technical analysis forms, the difference between opening and closing prices serves as an indicator of the direction and extent of intra-daily trends. They also noted that foreign exchange traders profit by accurately anticipating upward or downward trends and accordingly opening Long or Short positions. The relative size of the opening and closing prices also forms the basis of the strategy proposed in this article.

The novel hybrid approach we propose for interval-valued crude oil price forecasting can capture the nonlinear, dynamic features of interval-valued data while providing a good economic interpretation. Our method includes the variational mode decomposition (VMD), autoregressive conditional interval (ACI) model, and interval long short-term memory (iLSTM) model. Initially, we employ VMD to decompose the original time series into low- and high-frequency components, to represent the deterministic force for oil price evolution and normal market fluctuations respectively. Subsequently, we apply the autoregressive conditional interval (ACI) model to predict the low-frequency component by treating interval data as an inseparable random set, while the high-frequency component is predicted by interval long short-term memory (iLSTM) networks. Finally, we combine the two parts to generate interval-valued prediction. A novel interval-based trading strategy which obtains good financial performance is designed, additionally offering a valuable feature: Because its performance does not depend on any parameters, it is more reliable and robust than others. Empirical results show that the proposed forecasting method outperforms all benchmark models and the proposed trading strategy is superior to the compared benchmark strategies.

The proposed framework has some appealing features. First, by separating the high- and low-frequency components from the original series, we can apply appropriate models to predict these components according to their respective characteristics. To utilize the rich information contained in interval-valued oil price observations, we use the ACI model, which treats interval observation as an inseparable random set, to predict the low-frequency component which represents the deterministic force for oil price evolution. To better capture the nonlinear pattern, we apply the iLSTM model to forecast the high-frequency component, which is explained as normal market fluctuations. By separating components with different frequencies and applying appropriate models, prediction accuracy greatly improves.

That our paper is a first attempt to use the interval data composed of opening and closing prices is a second valuable aspect. By using interval data composed of opening and closing prices, we can effectively capture crude oil prices movements, allowing better use of historical

Table 1
Typical literature about crude oil price forecasting.

Ref.	Forecasting methods	Main results
Zhang et al. (2008)	Hybrid model: EEMD-ANN	The short-term fluctuations and long-term trends could be well forecasted by the ANN method.
Wei et al. (2010)	Single model: GARCH	There is not a model that wins in all forecasts, but overall the nonlinear GARCH model shows better forecasting power, especially for long-term forecast.
Movagharnjad et al. (2011)	Single model: NN	Neural network methods are able to enhance forecasting capability for all oil prices.
Jammazi and Aloui (2012)	Hybrid model: WD-BPNN	Predictions based on Wavelet denoising can lead to more realistic data projection with the minimum average percentage.
Xiang and Zhuang (2013)	Single model: ARIMA	Model ARIMA (1,1,1) possesses good prediction effect and can be used in short-term prediction of crude oil price.
Chiroma et al. (2015)	Hybrid model: GA-NN	The proposed GA-NN approach is better than the baseline algorithms in terms of prediction accuracy and computational efficiency.
Yu et al. (2017)	Single model: SVM	SVM model is a fairly good candidate for crude oil price forecasting in terms of either one-step prediction or multi-step prediction.
Sun et al. (2018)	Hybrid model: BEMD-iMLP-iHolt	The proposed forecasting method significantly outperforms other benchmark models at different forecasting horizons and different data frequencies.
Huang and Wang (2018)	Single model: RWNN	The empirical results demonstrate that the proposed RWNN model has a higher accuracy in crude oil price fluctuations predicting and is advantageous in improving the precision of prediction.
He et al. (2021)	Single model: ACI	ACI models outperform the popular point-based time series models.
Huang and Deng (2021)	Hybrid model: VMD-LSTM-MW	VMD-LSTM-MW model can improve prediction accuracy for both monthly and daily crude oil price prediction.
Drachal (2021)	Hybrid model: TVP-VAR	The Vector Autoregression approach results in more accurate forecasts than a single equation Time-Varying Regression or the standard Dynamic Model Averaging.
Zhou et al. (2022)	Hybrid model: NOFGHW	The new model can be used to predict monthly and quarterly energy time series, which is better than benchmark models. And the model demonstrates reliability in energy prediction.

Note: Ensemble empirical mode decomposition (EEMD); Artificial Neural Network (ANN); Generalized AutoRegressive Conditional Heteroskedasticity (GARCH); Neural Networks (NN); Wavelet decomposition (WD); Back propagation neural network (BPNN); Autoregressive Integrated Moving Average model (ARIMA); Genetic Algorithm (GA); Support Vector Machine (SVM); Bivariate empirical mode decomposition (BEMD); Interval multilayer perceptron (iMLP); Interval exponential smoothing method (iHolt); Random Wavelet Neural Network (RWNN); Autoregressive conditional interval (ACI); Variational Mode Decomposition (VMD); Long Short-Term Memory (LSTM); Moving-Window (MW); Time-Varying Vector Autoregression (TVP-VAR); New Optimized Fractional Grey Holt-Winters (NOFGHW).

information through the interaction of opening price and closing price. Notably, the directional insight provided by the interval data consisting of opening and closing prices is advantageous for both forecasting and trading endeavors. Furthermore, because we easily can place an order using opening or closing price, the disadvantage that interval data composed of high and low prices was not suitable for trading can be overcome.

This study, thirdly, contributes to the existing literature by offering a novel interval-based trading strategy for crude oil prices, one that, unlike previous interval prediction studies, does not only focus on prediction accuracy but also keeps sight of the importance of trading. Based on our proposed strategy, our proposed model obtains daily returns of 0.24 percent after transaction costs, a figure which is statistically and economically significant compared to competing models and strategies. Further analysis shows that the outstanding performance is robust to various forecasting horizons, specifically, before, during, and after the global COVID-19 outbreak.

The rest of the paper is organized as follows: Section 2 reviews the related literature. Section 3 introduces the related methodology and trading strategy. Section 4 presents empirical results and analysis. Section 5 provides an in-depth discussion of the financial performance of the proposed framework. Section 6 conducts robustness tests. Section 7 concludes.

2. Literature review

The abundant literature on developing reliable prediction methods to improve the prediction accuracy of crude oil prices can generally be divided into literature about single forecasting models and that about hybrid models; see Table 1.

2.1. Crude oil price forecasting: Single models

Single crude oil price forecasting methods can be roughly grouped into two categories: econometric models and artificial intelligence-based models.

Econometric models include the autoregressive integrated moving average (ARIMA) model (Xiang and Zhuang, 2013); the family of generalized autoregressive conditional heteroscedasticity (GARCH) models (Sadorsky, 2006; Cheong, 2009; Wei et al., 2010; Hou and Suardi, 2012; Efimova and Serletis, 2014); vector autoregression (VAR) models (Kilian, 2009) and error correction models (ECM) (Lanza et al., 2005). Such traditional econometric models have laid a solid foundation for the development of crude oil price forecasting (Huang and Deng, 2021). However, because they are unable to capture the nonlinear features of crude oil price time series (Bashiri Behmiri and Pires Manso, 2013), additional viable methods are being sought.

Artificial intelligence-based models, tapping into the contemporary, burgeoning field of AI research and application, have been introduced into the field of crude oil prediction to capture nonlinear features and improve prediction accuracy. Single nonlinear AI methods, including support vector machine (SVM) (Xie et al., 2006; Yu et al., 2017); decision tree (DT) method (Chen and He, 2019); neural networks (NN) (Movagharnjad et al., 2011; Yin and Wang, 2019), and long short-term memory (LSTM) networks (Gupta and Pandey, 2018) do achieve better forecasting performances than the econometric models. The drawbacks are that they suffer from issues of parameter optimization and overfitting (Sun et al., 2018). These artificial intelligence-based methods also face common challenges in that their prediction process is a “black box”, and they lack interpretability.

2.2. Crude oil price forecasting: Hybrid models

Widely used, hybrid models, are based on the principle of “divide and conquer” to predict crude oil prices and volatilities. These more-recent models decompose time series data into several intrinsic mode functions (IMFs). Each IMF then is meant to reflect a specified component having a specific feature, for example a linear or a nonlinear component, a component with high fluctuation, or white noise. To obtain more accurate predictions, an appropriate prediction method is then chosen according to the characteristics of each component. Because hybrid models can combine the merits of each model, they can be adapted to the complexities involved in predicting crude oil price (Yu et al., 2008; He et al., 2010b, 2012; Zhang et al., 2015; Yu et al., 2016; Zhao et al., 2017; Ding, 2018).

The performance of hybrid models is notably sensitive to the decomposition method (Yu and Ma, 2021). The four main kinds of decomposition methods are: Wavelet decomposition (WD) (Jammazi and Aloui, 2010, 2012); empirical mode decomposition (EMD) (Yu et al., 2008; Zhang et al., 2008, 2009; Xiong et al., 2013); ensemble empirical mode decomposition (EEMD) (Yu et al., 2016; Tang et al., 2018a,b), and variational mode decomposition (VMD) (Li et al., 2019, 2021b). Proposed by Walczak and Massart (1997), WD provides an efficient tool to analyze the hidden multiscale time-varying data patterns, but since the result of WD is sensitive to the basis function, an unstable performance may occur. Proposed by Huang et al. (1998), EMD does not need to preset the basis function. EMD decomposes time series into multiple inherent modal components according to the signal's time scale characteristics, which is free of supervision and has good adaptability. Subsequently, ensemble empirical mode decomposition (EEMD) is further developed to overcome the mode mixing problem in EMD (Wu and Huang, 2009). However, both EMD and EEMD are sensitive to noise and sampling which leads to a less robust result. To address this problem, Dragomiretskiy and Zosso (2013) proposed the VMD algorithm. VMD obtains effective components of different center frequencies in the frequency domain by the adaptive method, producing higher decomposition accuracy and more effective feature selection. With the advantage of adaptive and non-recursive traits, VMD is more suitable than other decomposition methods for crude oil price forecasting. Consistent with findings in Yu and Ma (2021) on VMD's suitability in the field, this paper applies VMD to decompose interval-valued crude oil prices.

VMD has attracted increasing attention in predicting crude oil price in the last few years. Jianwei et al. (2017) proposed a hybrid method combining VMD with independent component analysis (ICA) and ARIMA. With the acronym VMD-ICA-ARIMA, this hybrid model, when used to analyze and forecast the crude oil price, can generate more accurate prediction results. When Li et al. (2019) proposed hybrid models for monthly crude oil price forecasting using VMD along with artificial intelligence (AI) techniques, the results showed hybrid VMD-based models to be superior to previously popular EEMD-based models and single models in terms of both level and directional prediction accuracy. Li et al. (2021a) integrated VMD and random sparse Bayesian learning (VMD-RSBL), to forecast crude oil price. Their results demonstrate that the proposed VMD-RSBL model significantly outperforms many state-of-the-art schemes in terms of various evaluation indicators.

Although scholarly research on crude oil price prediction has had some breakthroughs, to date the literature in general tends to focus on predicting point-valued closing prices even though interval-based methods are likely to produce improved forecasting. This is because interval-valued data can cover more volatility information, such as uncertainty and variability (Sun et al., 2018). Within the field of interval prediction, the relevant literature usually treats interval data as two points, leading to insufficient use of information contained in the interval. A third limitation in the literature involves how the decomposition-ensemble model decomposes the time series into multiple IMFs (usually 7 or 8 IMFs). Therein, model complexity and calculation cost can increase greatly;

additionally, ensemble components prediction results may cause cumulative errors, which will cause deviations in the final result (Yu and Ma, 2021). Lastly, most existing research only focuses on prediction accuracy but ignores the importance of trading. Thus, it is highly desirable to develop a reliable trading strategy for interval-valued crude oil prices.

3. Methodology

In this section, the methodologies pertinent to the proposed prediction approach are delineated in detail, encompassing the VMD, ACI, and iLSTM techniques. Additionally, our interval-based trading strategy is introduced comprehensively.

3.1. Variational mode decomposition

Variational mode decomposition (VMD), as proposed by Dragomiretskiy and Zosso (2013), serves to decompose the original signal into a discrete number of sub-signals (modes) with the ability to properly cope with noise and adaptively determine the relevant bands. Dragomiretskiy and Zosso (2013) highlighted that VMD not only formulates the feature searching process as the optimization problem to identify the optimal risk factors but also proficiently addresses the well-known mode mixing problem. It generalizes the Wiener filter into a number of adaptive bands by decomposing a multicomponent input signal $f(t)$ into a set of quasi-orthogonal discrete band-limited subseries modes u_k , while reproducing the input signal $f(t)$ (Wang and Markert, 2016).

The typical VMD algorithm unfolds through several steps. Initially, the Hilbert transform is applied to the original data, yielding a unilateral frequency spectrum. Subsequently, an exponential tuned to the center frequency of the data is added to the mode's unilateral frequency spectrum to shift it to the baseband. Finally, the mode for the bandwidth is estimated utilizing $H1$ Gaussian smoothness of the demodulated data (Zou et al., 2020).

VMD decomposes the original signal $f(t)$ into the modes u_k by minimizing the bandwidth of each mode, which can be expressed in the form:

$$\min_{\{u_k\}, \{\omega_k\}} \left\{ \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) \otimes u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\}, \quad (1)$$

s.t. $\sum_k u_k = f(t),$

where u_k denotes subseries modes and ω_k denotes their respective central frequencies, k is the number of IMFs, $f(t)$ is the original input signal, $\delta(t)$ represents the Dirac distribution and $\otimes$ is the convolution operator.

VMD adopts a quadratic penalty term and a Lagrange multiplier to transform the constrained optimization problem into an unconstrained form. The augmented Lagrangian is defined as follows:

$$L(\{u_k\}, \{\omega_k\}, \lambda) = \alpha \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) \otimes u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 + \left\| f(t) - \sum_{k=1}^K u_k(t) \right\|_2^2 + \langle \lambda(t), f(t) - \sum_{k=1}^K u_k(t) \rangle \quad (2)$$

where α represents the balancing parameter of the data-fidelity constraint and λ represents the Lagrangian multiplier.

Finally, the solution to the above problem can be expressed as follows:

$$\hat{\mu}_k^{n+1} = \frac{\hat{f}(\omega) - \sum_{i \neq k} \hat{\mu}_i(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)^2},$$

$$\hat{\omega}_k^{n+1} = \frac{\int_0^\infty \omega |\hat{\mu}_k(\omega)|^2 d\omega}{\int_0^\infty |\hat{\mu}_k(\omega)|^2 d\omega}, \quad (3)$$

where n represents the number of iterations, and $\hat{f}(\omega)$, $\hat{\mu}_i(\omega)$, $\hat{\lambda}(\omega)$, and $\hat{\mu}_k^{n+1}$ represent the Fourier transforms of $f(t)$, $\mu_i(t)$, $\lambda(t)$, and $\mu_k^{n+1}(t)$ respectively.

3.2. Autoregressive conditional interval model

Autoregressive conditional interval (ACI) model, proposed by Han et al. (2016), can capture the conditional mean dynamics of an interval time series. The class of ACI models has also been extended to model an interval-valued vector process in Han et al. (2016). An ACI model with orders (p,q) is shown as follows:

$$X_t = \alpha_0 + \beta_0 I_0 + \sum_{j=1}^p \beta_j X_{t-j} + \sum_{j=1}^q \gamma_j u_{t-j} + u_t, \quad (4)$$

where X_t is an interval-valued time series, $\alpha_0, \beta_j (j = 0, \dots, p), \gamma_j (j = 1, \dots, q)$ are scalar parameters, $I_0 = [-\frac{1}{2}, \frac{1}{2}]$ is a unit interval and u_t is an interval martingale difference sequence (IMDS) with respect to the information set I_{t-1} , which means $E[u_t | I_{t-1}] = 0$ a.s.

It is noteworthy that X_t and u_t may allow the left bound of an ITS to be larger than the right bound, namely, extended random interval process. Specifically, $\{X_t\}$ is an extended random interval process if X_t on a probability space $(\Omega, \mathcal{F}, P)$ is a measurable mapping as $X_t(\omega) = [X_t^L(\omega), X_t^R(\omega)]$, where $X_t^L(\omega), X_t^R(\omega) \in \mathbf{R}$ for all $\omega \in \Omega$. We call it an extended random interval since we allow the case where $X_t^L > X_t^R$. In fact, the idea of extended interval has been considered in interval algebra by Kaucher (1980) as a so-called generalized interval. This extended interval can cover more applications in economics and finance. Additionally, the extended interval space ensures the completeness and consistency among the operation rules (Han et al., 2016).

Instead of using least squares estimation, Han et al. (2016) proposed that interval-based models should be estimated by minimizing D_K -distance. We use X, Y to represent extended interval $[x^L, x^R], [y^L, y^R]$ respectively. The D_K -distance between X and Y denoted by $D_K^2[X, Y]$ is defined as:

$$D_K^2[X, Y] = \begin{pmatrix} x^L - y^L \\ x^R - y^R \end{pmatrix}^T \begin{pmatrix} K_{LL} & K_{LR} \\ K_{RL} & K_{RR} \end{pmatrix} \begin{pmatrix} x^L - y^L \\ x^R - y^R \end{pmatrix}, \quad (5)$$

where $K = \begin{pmatrix} K_{LL} & K_{LR} \\ K_{RL} & K_{RR} \end{pmatrix}$ is a positive definite matrix. Based on the D_K -distance, the unknown parameters $\phi = (\alpha_0, \beta_0, \beta_1, \dots, \beta_p, \gamma_1, \dots, \gamma_q)'$ in model (4) can be estimated by the following equation:

$$\hat{\phi} = \underset{\phi \in \Phi}{\operatorname{argmin}} \hat{Q}_T(\phi), \quad (6)$$

where $\hat{Q}_T(\phi) = \sum_{t=1}^T D_K^2[X_t, \hat{X}_t(\phi)]$ is the sum of squared norm of residuals of the interval model.

The D_K metric considers the set of the absolute differences between all possible pairs of points (extreme and interior points) in extended intervals X and Y with a proper weighting function implied by matrix K . Thus, ACI model utilizes interval information more efficiently than conventional point-valued time series methods (He et al., 2021). Furthermore, by employing the estimation of D_K -distance, the ACI model exhibits robust performance in capturing the dynamics of interval data.

3.3. Long short-term memory for interval-valued time series

The long short-term memory (LSTM) network, proposed by Hochreiter and Schmidhuber (1997), is a specialized kind of recurrent neural network (RNN) that is designed to learn long-term rules. In contrast to traditional RNNs, which tend to update weights based on the most recent inputs during the gradient descent step of the training process, LSTMs introduce a gate mechanism to control the flow and preservation of information, effectively mitigating the vanishing gradient problem. Thus, it provides a mechanism for preserving longer memory in the model. A typical LSTM network comprises an input layer, output layer, and hidden layer. The hidden layer, distinctively, contains a forget gate, input gate, output gate, and a cell, with each gate controlling the inflow and outflow of information through the unit. Fig. 1 illustrates the structure of the hidden layer.

Let I_t denote the input extended interval of the hidden layer. In this context, $I_t = [D_t^L, D_t^R]$, in which D_t^L and D_t^R represent the high-frequency component of opening and closing prices respectively. Let h_{t-1} denote the output of the previous hidden layer; c_t represent the cell state of the hidden layer; and w_f, w_i, w_o, w_c symbolize the weights of forget gate, input gate, output gate and cell state layer, respectively. Additionally, f_t, i_t, o_t signify the forget rate, input rate and output rate, respectively; b_f, b_i, b_o, b_c denote the bias of the forget gate, input gate, output gate, and cell state layer; and $\sigma(\cdot)$ represents the sigmoid function. The computational processes at time t are as follows:

$$f_t = \sigma(w_f [h_{t-1}, I_t] + b_f), \quad (7)$$

$$i_t = \sigma(w_i [h_{t-1}, I_t] + b_i), \quad (8)$$

$$o_t = \sigma(w_o [h_{t-1}, I_t] + b_o), \quad (9)$$

$$c_t = f_t c_{t-1} + i_t \tanh(w_c [h_{t-1}, I_t] + b_c), \text{ and} \quad (10)$$

$$h_t = o_t \tanh(c_t). \quad (11)$$

These formulas determine the forget rate, input rate, output rate, cell state, and output of this hidden layer respectively.

3.4. Interval-based trading strategy

We propose a novel daily trading strategy, derived from the interval prediction of daily crude oil opening and closing prices. Following each trading session, we secure both the opening and closing prices along with their historical equivalents. We denote the one-step-ahead opening and closing price predictions of crude oil as $\hat{X}_t^L$ and $\hat{X}_t^R$, respectively, and use I_t to indicate the current position, with its status updated after the closure of each trading day. Here, $I_t = 0$ signifies holding cash without investment, while $I_t = 1$ indicates full investment in WTI crude oil futures. Adhering to Avellaneda and Lee (2010), transaction costs are set at 0.05% for each buying or selling action. We assume that short sales are permitted but leverage is not. Our strategy follows the following rules:

If $I_t = 0$ and $\hat{X}_{t+1}^L \leq \hat{X}_{t+1}^R$, we place a buy order at the next day's opening price, transitioning the position status, I_{t+1} , from 0 to 1. If $I_t = 0$ and $\hat{X}_{t+1}^L > \hat{X}_{t+1}^R$, a short position is opened at the following day's opening price and closed at its closing price, maintaining the position status I_{t+1} at 0. If $I_t = 1$ and $\hat{X}_{t+1}^L \leq \hat{X}_{t+1}^R$, the long position is held with no actions taken the following day, keeping I_{t+1} unchanged at 1. If $I_t = 1$ and $\hat{X}_{t+1}^L > \hat{X}_{t+1}^R$, all held crude oil is sold and a short position is opened at the following day's opening price, which is then closed at its closing price, altering the position status I_{t+1} from 1 to 0.

3.5. Proposed forecasting and trading framework

Our proposed framework consists of four main steps. Initially, variational mode decomposition (VMD) is deployed to decompose the original series into the IMF and the residual by setting the number of IMFs at $k = 1$. The IMF and the residual represent the low-frequency component and the high-frequency component of crude oil prices respectively.¹ Subsequently, we employ the autoregressive conditional interval (ACI) model to predict extended interval time series composed of the low-frequency components of opening and closing prices. Concurrently, We deploy the interval long short-term memory (iLSTM)

¹ Due to the intrinsic decomposition mechanism of VMD, the sum of low-frequency component and residual will slightly differ from the original series. We define the high-frequency component as the difference between original series and the IMF.

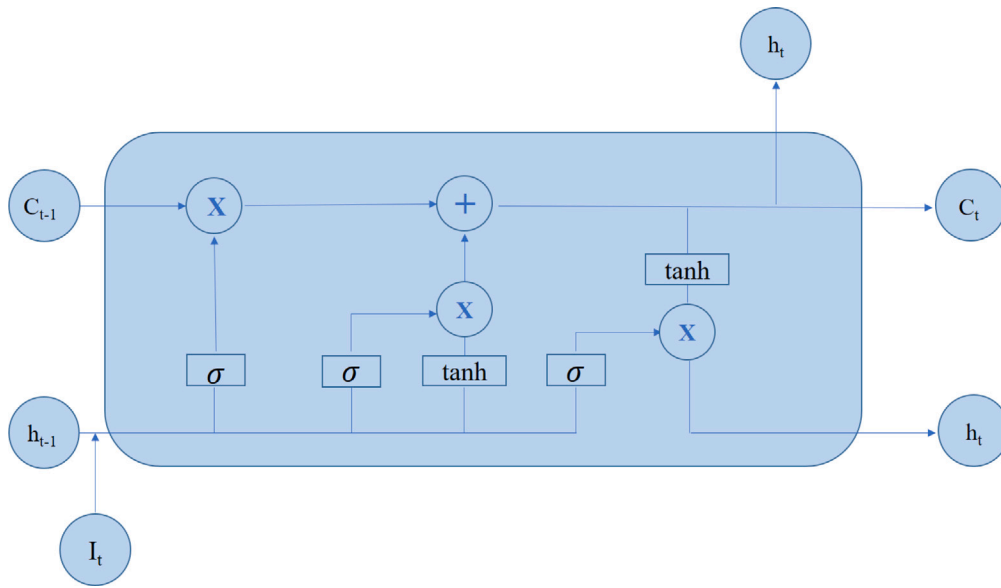


Fig. 1. The architecture of the iLSTM network.

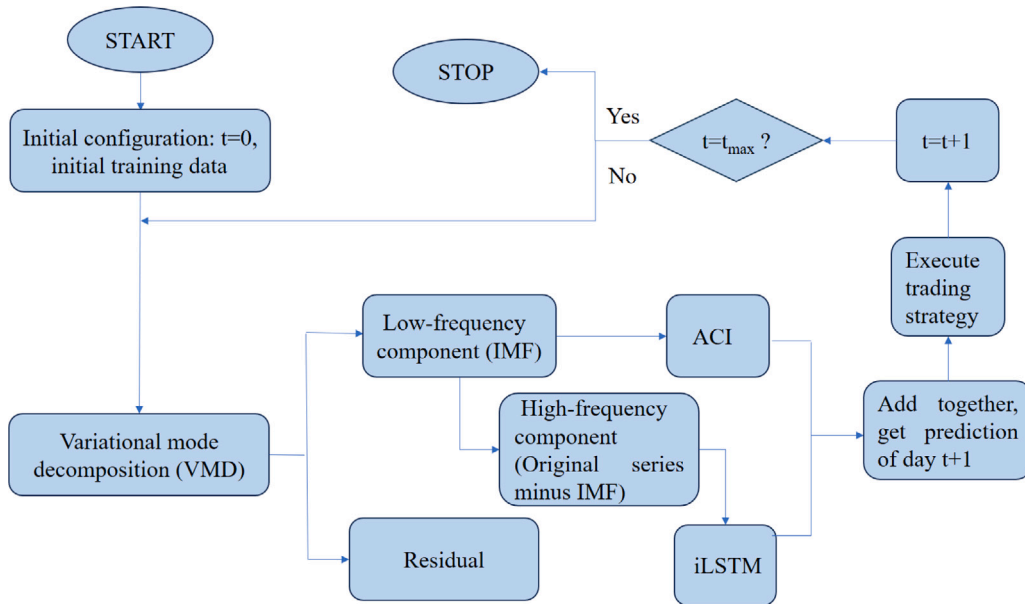


Fig. 2. Flow chart of the proposed framework.

network to predict extended interval time series composed of the high-frequency components of opening and closing prices. In the third step, we amalgamate the two parts to formulate the final interval-valued prediction. Ultimately, we implement corresponding trading strategies according to whether $I_t = 0$ and whether $\hat{X}_{t+1}^L > \hat{X}_{t+1}^R$. For each trading day, we repeat the aforementioned four steps. The framework is shown in Fig. 2.

4. Empirical results

4.1. Data description

We utilize daily interval-valued data of West Texas Intermediate (WTI) crude oil futures prices. For each day t , the left and right hands of an interval-valued oil price observation X_t are established using the opening and closing daily oil prices at day t .

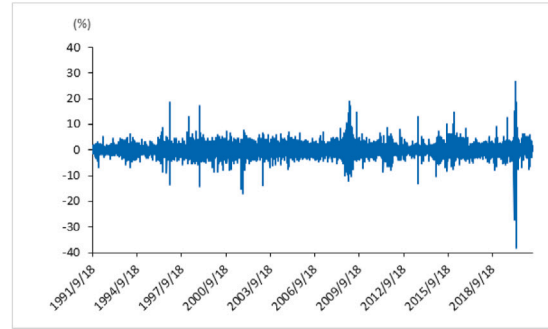
A rolling estimation scheme is adopted for a 6000 days rolling window. The first estimation sample spans the period from September 17, 1991, to July 9, 2015. We conduct one step-ahead out-of-sample forecasts for 1500 trade days, spanning from July 10, 2015, to June 7, 2021.

Fig. 3(a) presents the plots of the interval time series of WTI crude oil futures prices from September 17, 1991 to June 7, 2021. To examine the relationship between the variables, tests for the three common correlation coefficients are run for opening prices and closing prices: the Pearson correlation coefficient is 0.9993, the Kendall correlation coefficient is 0.9778, and the Spearman correlation coefficient is 0.9992. These results confirm a close connection between opening prices and closing prices.

Here, we calculate daily change by $r_t = \frac{C_t - C_{t-1}}{C_{t-1}}$, with C_t being the closing price on day t . Fig. 3(b) illustrates the daily change from July 10, 2015, to June 7, 2021. We find that the proportion of the absolute value of daily changes that are greater than 1% is 55.74%, and that the

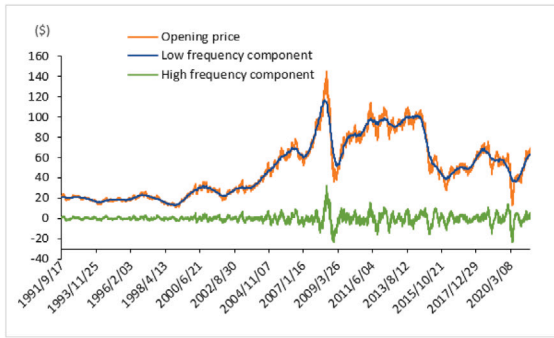


(a) Opening and closing price

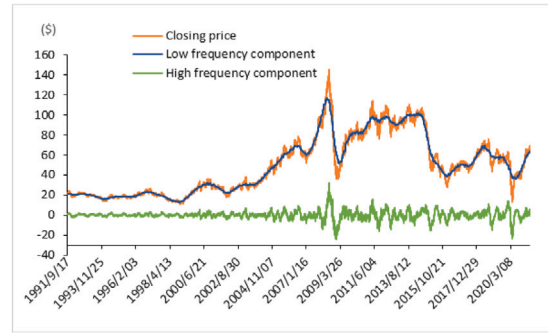


(b) Daily change

Fig. 3. Daily interval-valued time series and daily fluctuations in WTI crude oil futures.



(a) VMD results of opening prices



(b) VMD results of closing prices

Fig. 4. VMD results of opening and closing prices.

Table 2

The mean period and variance of each time series.

	MP	Variance		MP	Variance
Opening prices	3.9936	846.7239	Closing prices	3.9979	846.3121
Low-frequency component	22.1893	800.5088	Low-frequency component	22.4551	800.0411
High-frequency component	3.6692	19.6659	High-frequency component	3.6729	19.7110

proportion of those greater than 2% is 28.53%, which verifies the high volatility of crude oil prices.

We employ VMD to decompose the time series of opening and closing prices into the low-frequency component and the high-frequency component. While we apply VMD to the total data set here, aiming to better analyze the relationship between the original series and decomposed series, it is pertinent to note that VMD is only implemented on the training set during the forecasting process. This ensures avoidance of utilizing future information during forecasting. Fig. 4 displays the VMD results for opening and closing prices.

Following Zhang et al. (2008), we compute the mean period (MP) of both the low-frequency component and the high-frequency component. The Mean period is defined as the quotient obtained by dividing the total number of points by the number of peaks for each time series. Additionally, we calculate the variance of each time series. The results are shown in Table 2.

For the opening price series, the mean period is less than 4, indicating that the up or down trend of opening prices will change every 4 trading days. Using VMD, we obtain the low-frequency component with a mean period of 22.1893, which shows that the low-frequency component is steadier than the original series. The low-frequency component maintains a high correlation with the original price and accounts for more than 94% of variability, suggesting it is a deterministic force for oil price evolution. The high-frequency component proves to be more

volatile than the original series, since its mean period is 3.6692. Since the high-frequency component has no serious impact on oil price — it is generally within \$5, we see the high-frequency component represents normal market disequilibrium. The conclusion is similar for closing prices.

Table 3 presents the Pearson, Kendall, and Spearman correlation coefficients of opening (closing) prices, and the corresponding low-frequency component and high-frequency component. From Table 3, we observe a close relationship between opening (closing) prices and their low-frequency component, since all three correlation coefficients are all larger than 0.9. On the contrary, the high-frequency component has almost no connection with opening (closing) prices or with the low-frequency component. The finding supports the effectiveness of VMD in separating high- and low-frequency components.

4.2. Benchmark models

Our benchmark models come from both types: point-based models and interval-based models. Among the available point-based models, we choose autoregressive integrated moving average (ARIMA) model, generalized autoregressive conditional heteroskedasticity (GARCH) model, Holt exponential smoothing (HOLT) method, multilayer perceptron (MLP) model, and long-short term memory (LSTM) networks. For interval-based models, we employ the ACI model, MLP model

Table 3

Correlation coefficients.

Panel A. Pearson correlation coefficient (Opening price and its components)			
Correlation coefficient	Opening price	Low-frequency component	High-frequency component
Opening price	1.000000	0.988450	0.255271
Low-frequency component	0.988450	1.000000	0.105799
High-frequency component	0.255271	0.105799	1.000000
Panel B. Pearson correlation coefficient (Closing price and its components)			
Correlation coefficient	Closing price	Low-frequency component	High-frequency component
Closing price	1.000000	0.988418	0.252730
Low-frequency component	0.988418	1.000000	0.107815
High-frequency component	0.252730	0.107815	1.000000
Panel C. Kendall correlation coefficient (Opening price and its components)			
Correlation coefficient	Opening prices	Low-frequency component	High-frequency component
Opening price	1.000000	0.916737	0.128643
Low-frequency component	0.916737	1.000000	0.045281
High-frequency component	0.128643	0.045281	1.000000
Panel D. Kendall correlation coefficient (Closing price and its components)			
Correlation coefficient	Closing price	Low-frequency component	High-frequency component
Closing price	1.000000	0.916455	0.121643
Low-frequency component	0.916455	1.000000	0.042184
High-frequency component	0.121643	0.042184	1.000000
Panel E. Spearman correlation coefficient (Opening price and its components)			
Correlation coefficient	Opening price	Low-frequency component	High-frequency component
Opening price	1.000000	0.989249	0.175383
Low-frequency component	0.989249	1.000000	0.060412
High-frequency component	0.175383	0.060412	1.000000
Panel F. Spearman correlation coefficient (Closing price and its components)			
Correlation coefficient	Closing price	Low-frequency component	High-frequency component
Closing price	1.000000	0.989137	0.168462
Low-frequency component	0.989137	1.000000	0.057002
High-frequency component	0.168462	0.057002	1.000000

Table 4

Models specification.

Models	Environment	Function	Parameters
ARIMA	“forecast” package in R	auto.arima()	None
GARCH	“rugarch” package in R	ugarchfit()	Variance.model: model = sGARCH, garchOrder = c(1, 1), mean.model: armaOrder = c(1,1)
HOLT	R	HoltWinters()	None
ACI	Python	None	ACI(2,2) for data after first-order difference
MLP	“sklearn” in python	MLPRegressor()	Hidden layer sizes = (100,50), activation = relu, solver = adam, alpha = 0.01, max iter = 200
iMLP	“sklearn” in python	MLPRegressor()	Hidden layer sizes = (100,50), activation = relu, solver = adam, alpha = 0.01, max iter = 200
LSTM	“sklearn” in python	None	Hidden number = 20, optimizer = adam, loss = mean squared error, learning rate = 0.001
iLSTM	“sklearn” in python	None	Hidden number = 20, optimizer = adam, loss = mean squared error, learning rate = 0.001

for interval (iMLP), and LSTM network for interval (iLSTM) to predict opening and closing prices simultaneously. One useful difference between MLP (LSTM) and iMLP (iLSTM) is that the former predicts opening and closing prices respectively, while the latter predicts them simultaneously.

We use ACI (2,2) to predict the interval data obtained after the first-order difference because (1) the first-order difference of opening and closing prices are stationary, and (2) ARMA (2,2) is appropriate for both of them. For the same reason, we also use ACI (2,2) to predict the low-frequency component interval data when we employ the VMD-ACI-iLSTM model. As for the matrix K in D_k -distance, we set $K_{LL} = K_{RR} = \frac{1}{3}$, $K_{LR} = K_{RL} = \frac{1}{6}$ here because this special case is the same as the symbolic covariance method (SCM) proposed in Billard (2007) and in Xu (2010). For predicting interval data composed of high-frequency components of opening and closing prices, our iLSTM models contain an input layer with 6000 timesteps, and 2 features and 50 hidden layers; the optimizer is “adam”, the loss is “mean squared error”, and we set the learning rate to be 0.001. See further details in Table 4.

4.3. Forecast criteria

We employ six criteria to fully evaluate the forecast performance of different methods. First, the interval U of Theil statistics (U^I) and the

interval average relative variance (ARV^I) are applied to compare the forecasting errors of the reference model and a naïve model (Maia and de Carvalho, 2011); they are defined as follows:

$$U^I = \sqrt{\frac{\sum_t (X_{t+1}^R - \hat{X}_{t+1}^R)^2 + \sum_t (X_{t+1}^L - \hat{X}_{t+1}^L)^2}{\sum_t (X_{t+1}^R - X_t^R)^2 + \sum_t (X_{t+1}^L - X_t^L)^2}}, \quad (12)$$

$$ARV^I = \frac{\sum_t (X_{t+1}^R - \hat{X}_{t+1}^R)^2 + \sum_t (X_{t+1}^L - \hat{X}_{t+1}^L)^2}{\sum_t (X_{t+1}^R - \bar{X}^R)^2 + \sum_t (X_{t+1}^L - \bar{X}^L)^2}. \quad (13)$$

U^I is first used to compare the forecast errors of the reference model with the random walk. $U^I < 1$ implies that the reference model outperforms random walk, while $U^I > 1$ implies that the reference model underperforms random walk. When U^I tends toward zero, the reference model is perfect. ARV^I is used to compare the reference model's forecast errors with the time series mean. $ARV^I < 1$ indicates that the reference model performs better and $ARV^I > 1$ indicates that the time series mean performs better. An ARV^I of 0 indicates that the reference model is perfect.

Second, the interval mean square error (MSE^I), the interval mean distance error (MDE^I) and the interval mean absolute percentage

Table 5
Performance comparisons.

Model	U^I	ARV^I	MSE^I	MDE^I	$MAPE^I$	DA
ARIMA	1.0036	0.1127	1.2238	0.8734	0.01913	0.4900
GARCH	1.0160	0.1140	1.2389	0.8898	0.01953	0.4867
HOLT	1.0068	0.1129	1.2277	0.8742	0.01915	0.4980
MLP	1.3099	0.1469	1.5957	1.1655	0.02563	0.5040
LSTM	1.0111	0.1134	1.2331	0.8737	0.01923	0.4906
ACI	1.0366	0.1163	1.2641	0.9023	0.01970	0.4833
iMLP	0.9943	0.1115	1.1914	0.8442	0.01851	0.4973
iLSTM	0.9814	0.1101	1.1650	0.8259	0.01809	0.4933
VMD-ACI-iLSTM	0.8421	0.0945	0.9615	0.6388	0.01457	0.5167

error ($MAPE^I$) are used to fully compare the prediction accuracy of different models; they are defined as follows:

$$MSE^I = \frac{1}{2} \left(\sqrt{\frac{\sum_i (X_t^R - \hat{X}_t^R)^2}{n}} + \sqrt{\frac{\sum_i (X_t^L - \hat{X}_t^L)^2}{n}} \right), \quad (14)$$

$$MDE^I = \frac{1}{2} \cdot \sum_i \frac{1}{n} (|X_t^R - \hat{X}_t^R| + |X_t^L - \hat{X}_t^L|), \quad (15)$$

and

$$MAPE^I = \frac{1}{2n} \left(\sum_i \left| \frac{X_t^R - \hat{X}_t^R}{X_t^R} \right| + \sum_i \left| \frac{X_t^L - \hat{X}_t^L}{X_t^L} \right| \right). \quad (16)$$

For each of these three criteria, a smaller value implies a better interval forecast.

Third, to be pragmatic in the actual financial realm, a proficient prediction model necessitates not only robust fitting performance but also superior predictive power. Therefore, we introduce directional accuracy (DA) to evaluate the predictability of the model. DA is defined as:

$$DA = \sum_{i=1}^n a_i / n, \quad (17)$$

where

$$a_i = \begin{cases} 1 & (X_t^L - X_t^R) (\hat{X}_t^L - \hat{X}_t^R) > 0, \\ 0 & \text{otherwise.} \end{cases}$$

The larger the DA , the higher the predictive ability of the model.

To further verify the superiority of the proposed framework, we conduct the Diebold–Mariano (DM) test to evaluate whether there is significant difference between two models. See details in [Diebold and Mariano \(2002\)](#).

4.4. Prediction results analysis

[Table 5](#) provides the evaluation results and [Table 6](#) furnishes the results from the Diebold–Mariano (DM) test. Through our analysis, we deduce the following pivotal findings:

Firstly, our proposed VMD-ACI-iLSTM model yields the most favorable forecasting results under all criteria. It attains the minimum prediction error and the maximum directional accuracy. Additionally, as seen in [Table 6](#), the exemplary performance of the VMD-ACI-iLSTM model is statistically significant under the 0.01 significance level when compared to other models. This might be attributable to the model treating an interval observation as a set and directly modeling the interval data, thereby efficiently utilizing the rich information contained within interval data. Another plausible explanation is that through data decomposition and by assigning appropriate models to different components based on their respective data characteristics, the VMD-ACI-iLSTM model enables more precise predictions.

Secondly, it is discernible that interval models generally outperform point models in the majority of cases benefitting from the rich information contained in interval data. In particular, the iLSTM model and iMLP model significantly surpass the LSTM model and MLP model in performance, affirming that the information embedded in interval data is advantageous for prediction.

Thirdly, from the index U^I , it is observable that none of the employed point-valued prediction models (ARIMA, GARCH, HOLT, MLP, and LSTM) can yield better predictions than random walk, which aligns with findings in [He et al. \(2021\)](#). This may stem from the fact that crude oil prices, characterized by their high volatility, are challenging to predict—a phenomenon also evident in exchange rate predictions. [Meese and Rogoff \(1983\)](#) reported that no model can obtain better prediction results than random walk persistently. However, by using the information contained in interval data, we ascertain that, in most cases (excluding the ACI model), the interval-valued prediction methods secure better results than random walk. This further solidifies the assertion regarding the potent predictive capacity of interval data.

Fourthly, it is worth noting that the DA index reveals that the VMD-ACI-iLSTM model achieves the highest accuracy in directional prediction, a critical factor as it lays the foundation for the success of our trading strategy, as proposed in [Section 3](#). Furthermore, it is observable that a slight increase in DA can catalyze an explosive growth in yields. These results are robust to various forecasting periods.

5. Financial performance

5.1. Benchmark strategy

We compare the return characteristics of eight benchmark models and our proposed VMD-ACI-iLSTM model using the designed strategy to investigate each model's financial performance. For our strategy benchmarks, buy and hold (BH) strategy, moving average (MA) strategy and relative strength index (RSI) strategy are chosen to further prove the superiority of our proposed framework.

BH strategy means that we buy crude oil at the closing price of the first trading day and hold it for the duration of the trading period. The MA strategy, a classical trend tracking strategies, stipulates that for day t with closing price p_t , we compute $MA(t)$ with:

$$MA(t) = \sum_{i=t-m_1+1}^t \frac{p_i}{n}, \quad (18)$$

where m_1 is a predefined parameter. If $I_t = 0$ and $P_t > MA(t)$, it represents that a short-term upward trend has been formed and we buy crude oil at price p_t ; if $I_t = 1$ and $P_t < MA(t)$, it represents that a short-term downward trend has been formed and we sell crude oil at price p_t ; otherwise, no action is taken.

The relative strength index (RSI) analyzes the intention and strength of buying and selling in the market by comparing the average increase and decrease in closing prices over a period. $RSI(t)$ is defined as:

$$RSI(t) = \frac{A}{A+B}, \quad (19)$$

where $A = \sum_{i=t-m_2+1}^t p_i I(p_i > 0)$, $B = \sum_{i=t-m_2+1}^t p_i I(p_i < 0)$ and m_2 is a predefined parameter. Here, $I = 1$ if the condition in parentheses is met and $I = 0$ if not.

Rules of RSI strategy are: if $I_t = 0$ and $RSI(t) < 0.4$, it indicates that the crude oil is oversold, so we buy crude oil at price p_t ; if $I_t = 1$ and $RSI(t) > 0.7$, it indicates that the crude oil is overbought, so we sell crude oil at price p_t ; otherwise, there is no clear trend and no action is taken.

Obviously, m_1 in MA strategy and m_2 in RSI strategy are important for determining the performance of the strategies. To make the comparison results more reliable and robust, we explore values of m_1 from 2 to 20 and m_2 from 10 to 100. The optimal performers – MA strategy with $m_1 = 8$ and RSI strategy with $m_2 = 68$ – are utilized as the benchmark strategies, dubbed MA* and RSI*, respectively.

Table 6

DM test results.

Panel A. Opening price								
	ARIMA	GARCH	HOLT	MLP	LSTM	ACI	iMLP	iLSTM
GARCH	−2.3347*							
HOLT	−1.2066	1.0569						
MLP	−7.5266**	−6.4082**	−6.7877**					
LSTM	−1.3263	1.3232	0.3570	7.6386**				
ACI	−3.7490**	−1.1370	−2.1056*	8.4592**	−3.5566**			
iMLP	10.9821**	10.3113**	10.9708**	13.4922**	10.9556**	13.6434**		
iLSTM	12.4822**	12.0962**	12.8549**	12.8909**	12.2462**	13.3797**	5.2566**	
VMD-ACI-iLSTM	13.2684**	13.6483**	13.7363**	13.0581**	12.8025**	13.0595**	−9.8278**	−9.3027**
Panel B. Closing price								
	ARIMA	GARCH	HOLT	MLP	LSTM	ACI	iMLP	iLSTM
GARCH	−0.9954							
HOLT	0.4541	1.2186						
MLP	−10.4475**	−9.7007**	−10.4450**					
LSTM	−1.3299	−0.2544	−1.4036	11.1842**				
ACI	−3.1213**	−1.5824	−3.1825**	11.1882**	−3.6083**			
iMLP	−6.6817**	−5.4128**	−6.5872**	10.1601**	−7.6786**	−7.8628**		
iLSTM	−7.2859**	−6.0218**	−7.2599**	8.6078**	−8.9183**	−8.7778**	−1.4783	
VMD-ACI-iLSTM	−4.804**	−3.8817**	−4.6274**	9.4154**	−5.1048**	3.4078**	−3.6128**	−4.4937**

* Denotes the result is significant under 0.05 significance level. ** denotes the result is significant under 0.01 significance level.

Table 7

Comparisons between different prediction methods and trading strategies.

	ARIMA	GARCH	HOLT	MLP	LSTM	ACI	iMLP	iLSTM	VMD-ACI-iLSTM	BH	MA*	RSI*
Cumulative profits (long %)	42.40	5.96	63.01	91.27	53.50	16.28	120.29	60.89	353.12			
Cumulative profits (short %)	75.76	38.00	107.71	127.52	94.88	38.28	136.13	18.47	278.42			
Total profits (%)	150.28	46.22	238.58	335.18	199.14	60.81	420.20	90.60	1615.43	29.79	90.38	61.23
Mean daily return (%)	0.10	0.07	0.13	0.14	0.12	0.08	0.15	0.09	0.24	0.06	0.06	0.07
Minimum (%)	−12.89	−12.89	−12.89	−15.86	−12.89	−15.86	−15.86	−14.49	−15.59	−38.17	−15.63	−38.17
Maximum (%)	62.41	62.41	62.41	62.41	62.41	62.41	62.41	62.41	63.10	26.80	17.17	26.80
Annualized return (%)	16.66	6.59	22.73	28.02	20.21	8.31	31.92	11.44	61.19	4.49	11.42	8.35
Max. drawdown (%)	52.15	58.54	53.46	34.27	51.62	59.22	43.86	75.53	38.17	82.78	43.79	72.39
Sharpe ratio	0.4696	0.2901	0.5706	0.6591	0.5293	0.3220	0.7184	0.3791	1.0534	0.2405	0.3489	0.2981
Sortino ratio	0.8383	0.5101	1.0255	1.1604	0.9471	0.5574	1.2993	0.6968	2.0415	0.3829	0.6031	0.4716
Number of transactions	1078	1122	1114	885	1083	1018	829	1498	891	1	148	3

5.2. Results analysis

Comparative results of financial performance are shown in Table 7, and the yield curves are presented in Fig. 5.

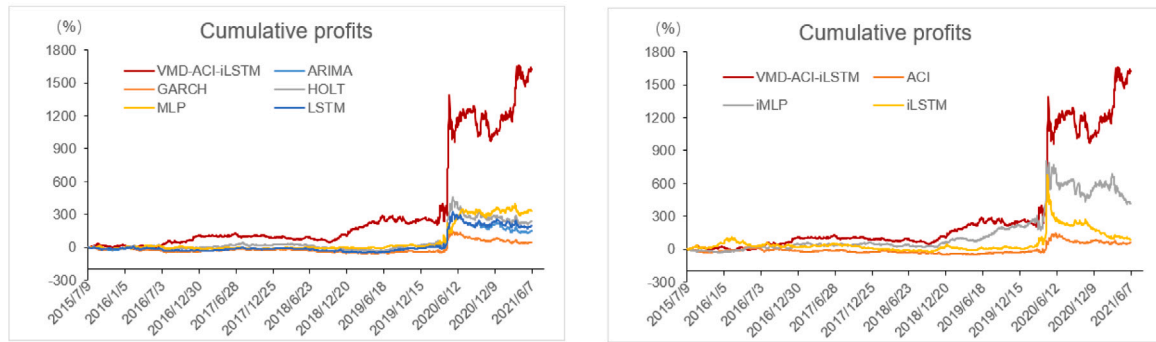
In Table 7, we present 11 indexes, dividing them into four categories: return characteristics (which include cumulative profits from long positions, cumulative profits from short positions, total profits, mean daily return, minimum, maximum and annualized return); risk characteristics (including maximum drawdown); risk-adjusted return characteristics (including Sharpe ratio and Sortino ratio); and trading frequency characteristics (including number of transactions). The following are key findings from four perspectives:

(1) Regarding return characteristics: The VMD-ACI-iLSTM model exhibits significant advantages compared with the performance of the eight benchmark prediction models using our proposed trading strategy. Its returns from going long on crude oil were 353.12%, which are almost three times that of the second place. The returns from shorting crude oil were 278.42%, which are more than twice that of the second place. VMD-ACI-iLSTM obtains total profits of 1615.43%, far exceeding the performance of the second place, i.e., iMLP. The same happens when mean daily return and annualized return are compared: VMD-ACI-iLSTM significantly outperforms other benchmark models. When using our proposed trading strategy, most of the eight prediction models show immense superiority compared to the three benchmark strategies (BH, MA* and RSI*). BH strategy is the worst among all prediction methods and trading strategies with only 29.79% total profits. The performance of MA* is only slightly better than that of GARCH and ACI, and MA* is at a significant disadvantage compared to the other prediction methods. These results confirm the superiority of our proposed VMD-ACI-iLSTM model and trading strategy.

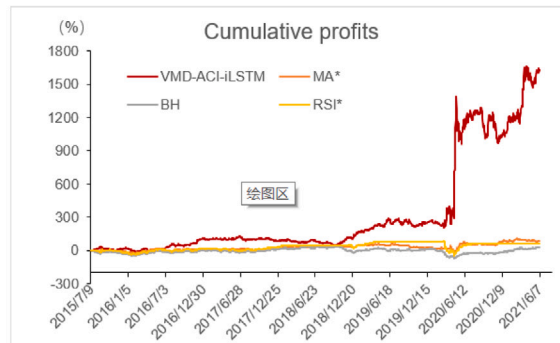
We observed that BH and RSI* strategies got the same minimum daily return of −38.17%. Both reached their minimum return on April 21, 2020, when crude oil price fell from \$21.22 per barrel to \$13.12 per barrel due to financial panic caused by COVID-19. In fact, before April 21, 2020, the closing prices of crude oil had been lower than the opening prices for two consecutive days; thus, all the interval and point models correctly predicted that on April 21, 2020, the closing price will be lower than the opening price. All models not only avoided the sharp drop but also had a daily return of 62.41% by the shorting mechanism. Although they have the maximum daily return on the same day, the maximum daily return of VMD-ACI-iLSTM is slightly higher than that of other models. Because VMD-ACI-iLSTM uses VMD to improve prediction, there is a slight prediction difference between VMD-ACI-iLSTM and other models. After the closing of the crude oil market on April 20, 2020, we remained long in crude oil under the VMD-ACI-iLSTM model, then sold crude oil at the opening price on April 21, 2020, and shorted it. By way of contrast, when we sold all crude oil on April 20, 2020, under other models, we only got the returns of shorting oil on April 21, 2020. As for MA*, since the closing price was below the 8-day moving average line on April 20, 2020, we had sold all crude oil according to MA* strategy, avoiding the drop on April 21, 2020, too. However, since RSI* uses oil price performance of the past 68 days, it was less sensitive than MA* strategy and failed to respond to the drop on April 21, 2020.

(2) Regarding risk characteristics: VMD-ACI-iLSTM achieved a competitive maximum drawdown of 38.17%, a figure better than all models and strategies in this study except MLP, which was 34.27%.

(3) Regarding risk-adjusted return characteristics: The Sharpe ratio measures how much excess return an investment generates by taking one unit of total risk. The Sortino ratio is similar to Sharpe ratio but,



(a) Performance of VMD-ACI-iLSTM and point-valued methods (b) Performance of VMD-ACI-iLSTM and other interval-valued methods



(c) Performance of VMD-ACI-iLSTM and benchmark trading strategies

Fig. 5. Performance of cumulative profits.

to distinguish adverse and favorable fluctuations, Sortino ratio uses the downward biased standard deviation rather than the total standard deviation. A higher Sortino ratio indicates that one can obtain higher excess returns by taking one unit of the downside risk. The results reveal consistency between the Sharpe ratio and Sortino ratio: the higher the Sharpe ratio, the higher the Sortino ratio. The VMD-ACI-iLSTM model achieves the highest Sharpe ratio, 1.0534, with iMLP trailing in second at a mere 0.7184. Similar results were observed for the Sortino ratio.

(4) Regarding trading frequency characteristics: Firstly, the number of transactions associated with the BH, MA* and RSI* strategies is markedly lower compared to the strategy we propose. MA* executes more trades than RSI* because MA* is more sensitive to price changes than RSI*; secondly, the VMD-ACI-iLSTM model is categorized within the low-transaction group, encompassing MLP, iMLP, and VMD-ACI-iLSTM. The low number of transactions is also an important reason for their outstanding performance since we impose a transaction fee of 0.05% every time we buy or sell.

In conclusion, two key findings emerge: firstly, when compared to existing strategies, i.e., BH, MA and RSI, our proposed strategy demonstrates superior performance across most prediction methods; secondly, among all prediction methods, the VMD-ACI-iLSTM model obtains the highest return, the smallest maximum drawdown, the highest risk-adjusted return and a low number of transactions. These findings underscore the superiority of our proposed framework.

5.3. A critical review over time

Fig. 5 illustrates the performance of different models and benchmark strategies over time, i.e., from July 10, 2015, to June 7, 2021. It is

observable that for all prediction methods utilizing the strategy we propose, cumulative profits underwent explosive growth between February 1, 2020 and May 31, 2020, a period during which the crude oil market experienced shock due to COVID-19. In order to provide further insight into this phenomenon, we focus on four most competitive methods with respect to the strategy performance, i.e., MLP, iMLP, iLSTM, and VMD-ACI-iLSTM. Although the performance of iLSTM model is not outstanding, we include it in the comparison given its superior performance during the epidemic. We divide the forecast horizon into three periods as follows, and in Figs. 6 and 7, we display their profit performance, while Table 8 shows their prediction errors.

During the period from July 10, 2015, to January 31, 2020, the crude oil market had not yet been affected by the epidemic. The crude oil market experienced wide shocks and the closing price shifted from \$53.28 to \$51.63 per barrel during this period. At this stage, VMD-ACI-iLSTM delivered the best performance with average daily return of 0.1312% and a directional accuracy of 51.04%. Interestingly, we found iMLP to be slightly weaker than VMD-ACI-iLSTM, and they exhibited similarly good performance during this period, as seen in Fig. 6(a). In sharp contrast, MLP and iLSTM yielded almost no gains after transaction costs. The directional accuracy of iLSTM was the only one less than 50% which led to its subpar financial performance. Regarding the Sharpe ratio and Sortino ratio, it is evident that iMLP and VMD-ACI-iLSTM are the top two with comparable performance.

During the period from February 1, 2020, to May 31, 2020, the crude oil market was severely impacted by the epidemic but then gradually recovered. The closing price of WTI crude oil futures showed clear trends, plummeting from \$51.63 to \$13.12 per barrel and then rebounding to \$35.52 per barrel. During the time, the price underwent significant increases and decreases. For example, it plunged by 27.26% on March 9, 2020, soared by 26.8% on April 17, 2020, and collapsed

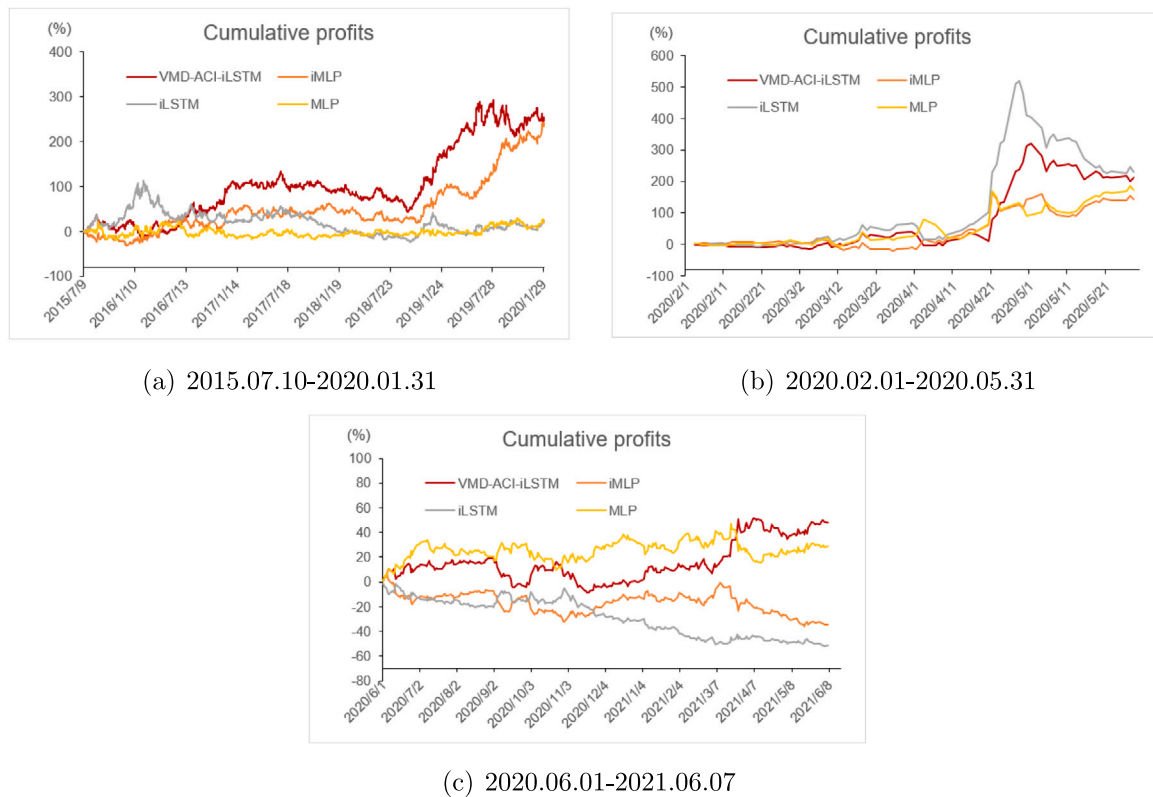


Fig. 6. Comparisons of different time periods.

by 38.17% on April 21, 2020. Due to the clear upward and downward trends, cumulative profits displayed explosive growth for all prediction methods. Among the four prediction methods, iLSTM demonstrated remarkable advantages with a directional accuracy of 59.76% and an average daily return of 1.9727%. It is reasonable to perceive iLSTM as inherently prone to overfitting, which explains why iLSTM performs well when there is a clear up or down trend but struggles when prices fluctuate violently around a certain value. However, due to the unstable performance of iLSTM, it only exhibits a weak advantage over VMD-ACI-iLSTM, with iLSTM achieving total profits of 230.04% and VMD-ACI-iLSTM securing 212.19%.

During the period from June 1, 2020, to June 7, 2021, the crude oil market managed to shake off the epidemic's impact and rebound to pre-epidemic levels. The closing price rose from \$35.52 to \$69.19 per barrel. During the time, both iLSTM and iMLP achieved negative total returns, which are shown in Fig. 6(c). Once again, VMD-ACI-iLSTM proved to be superior. It not only secured the best positive returns but also exhibited optimal performance in directional accuracy, Sharpe ratio, and Sortino ratio.

It is worth noting that, across these three time periods, the proposed VMD-ACI-iLSTM model consistently outperforms on the majority of error comparison metrics, underscoring the robustness of our forecasting model across the temporal dimension.

In summary, we distill two key findings. First, the VMD-ACI-iLSTM model demonstrates steady performance, benefiting from the separation of high- and low-frequency components, and securing positive returns before, during, and after the global outbreak of COVID-19. Second, the iLSTM potentially exhibits an overfitting problem. When the market displays clear trends, iLSTM excels as it adeptly learns the trend. However, when the market undergoes severe fluctuations, iLSTM struggles to produce accurate directional predictions due to overfitting, resulting in subpar performance. In contrast, VMD-ACI-iLSTM manages to generate significant positive returns, withstanding market fluctuations.

Table 8

Performance in different time periods.

Panel A. 2015.07.10–2020.01.31						
	U^I	ARV^I	MSE^I	MDE^I	$MAPE^I$	DA
MLP	1.3139	0.1499	1.4218	1.1058	0.02184	0.5034
iMLP	0.9818	0.1119	1.0390	0.7912	0.01563	0.5061
iLSTM	0.9830	0.1121	1.0314	0.7913	0.01559	0.4983
VMD-ACI-iLSTM	0.8080	0.0921	0.7947	0.5840	0.01161	0.5104
Panel B. 2020.02.01–2020.05.31						
	U^I	ARV^I	MSE^I	MDE^I	$MAPE^I$	DA
MLP	1.3674	0.2889	3.4919	2.4466	0.09267	0.5731
iMLP	1.0525	0.2224	2.6701	1.7290	0.06568	0.5122
iLSTM	1.0368	0.2191	2.6141	1.6830	0.06413	0.5976
VMD-ACI-iLSTM	0.9729	0.2055	2.4332	1.6311	0.06551	0.5609
Panel C. 2020.06.01–2021.06.07						
	U^I	ARV^I	MSE^I	MDE^I	$MAPE^I$	DA
MLP	1.2021	0.1306	1.3570	1.0279	0.02140	0.5076
iMLP	0.9531	0.1035	1.0569	0.8018	0.01649	0.4885
iLSTM	0.8839	0.0960	0.9608	0.7105	0.01475	0.4389
VMD-ACI-iLSTM	0.7501	0.0814	0.7404	0.5681	0.01167	0.5305

6. Robustness

In this section, we undertake additional empirical analyses to affirm the robustness of proposed VMD-ACI-iLSTM model. Specifically, we integrate sentiment information and trend information into the proposed VMD-ACI-iLSTM model, and scrutinize the prediction errors and profit performance.

6.1. Sentiment data

We utilized the CBOE Crude Oil Volatility Index (OVX) as a measure of market sentiment. OVX is a sentiment-driven index that tracks the

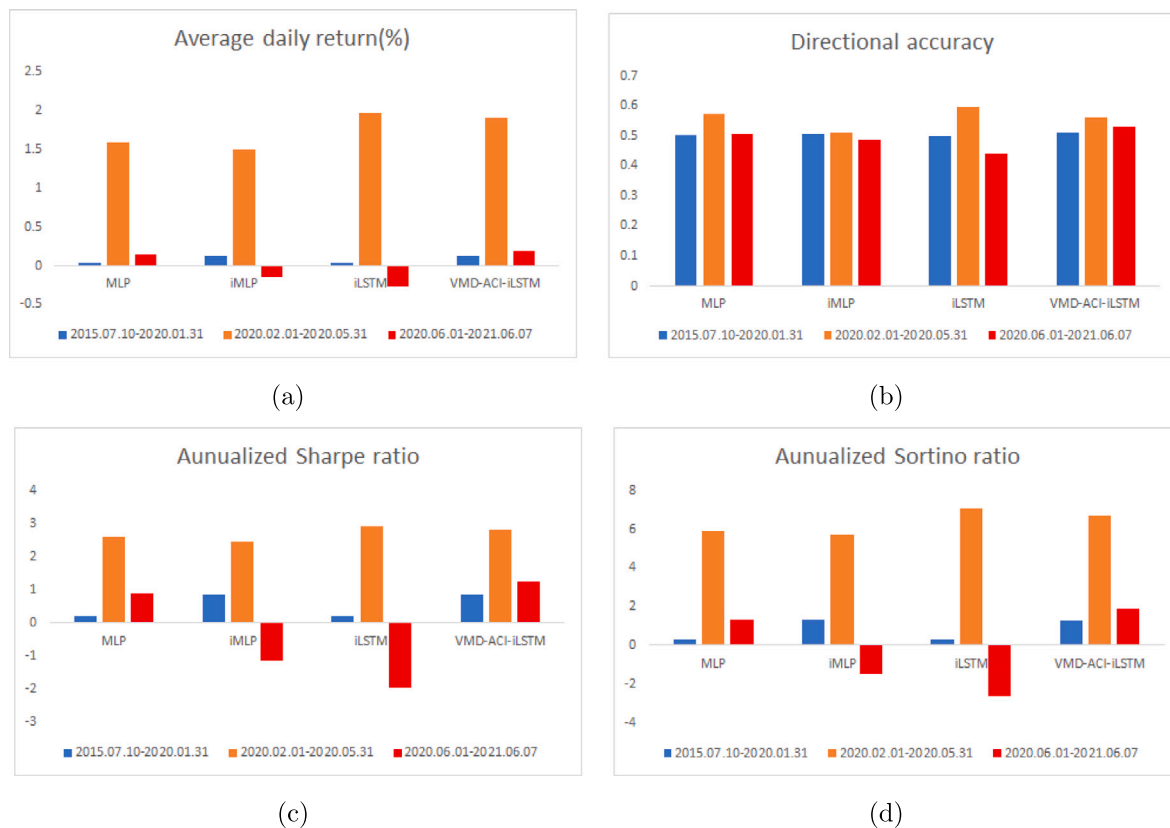


Fig. 7. Comparison of MLP, iMLP, iLSTM and VMD-ACI-iLSTM before, during and after the global COVID-19 epidemic.

Table 9

Comparison of the VMD-ACI-iLSTM model with and without sentiment data.

	U^I	ARV^I	MSE^I	MDE^I	$MAPE^I$	DA	AR
Original Model	0.8421	0.0945	0.9615	0.6388	0.01457	0.5167	61.19%
New Model	0.8870	0.0995	1.0294	0.6607	0.01560	0.5186	61.51%

“Original Model” represents the VMD-ACI-iLSTM model, while “New Model” represents the VMD-ACI-iLSTM model with sentiment data incorporated. “AR” is an abbreviation for “Annualized Return”.

crude oil price volatility as priced by the United States Oil Fund (USO) and reflects investors’ expectations of crude oil price volatility over the next 30 days. Hence, it serves as a good proxy for market sentiment and risk perception in the crude oil market. [Bossman et al. \(2023\)](#) examined the asymmetric relationships between EU sectoral stocks and OVX. They found that, similarly to the CBOE Volatility Index (VIX), OVX also possesses some hedging and safe-haven attributes against EU stocks. When market participants anticipate higher volatility in future WTI prices, USO option prices increase, resulting in an increase in OVX. Conversely, OVX decreases when market expectations of volatility decline.

Since we decompose the crude oil data using Variational Mode Decomposition (VMD), the low-frequency component represents the fundamental value, which is not influenced by short-term trader sentiment. The residual term, on the other hand, captures short-term fluctuations which is affected by investor sentiment. Therefore, we incorporated OVX data into the prediction of the residual term to obtain the final forecasting results. The performance of various evaluation metrics after incorporating OVX data can be seen in [Table 9](#).

Based on [Table 9](#), several observations can be made. First, U^I , ARV^I , MSE^I , MDE^I and $MAPE^I$ all increased and these results are statistically significant at the 0.05 significance level, indicating

that the introduction of OVX emotional data in our decomposition forecasting framework may not be advantageous for price prediction. We believe this could be attributed to the fact that the residual term after VMD decomposition already captures high-frequency fluctuations influenced by various factors including emotional data. In this scenario, incorporating additional emotional data from traders might amplify the noise rather than improve the prediction accuracy. Second, the directional accuracy (DA) slightly improved, suggesting that emotional data has a positive impact on directional forecasting, thereby enhancing strategy performance.

These findings indicate that while the inclusion of emotional data might not directly improve the overall prediction accuracy, it can be beneficial in terms of enhancing the accuracy of directional predictions and, consequently, improving strategy performance. The inability of incorporating sentiment data to enhance the prediction accuracy of the VMD-ACI-iLSTM model provides additional evidence of the effectiveness of VMD decomposition.

6.2. Trend data

In the analysis of [Section 5.3](#), we observe that iLSTM effectively learns trends, resulting in favorable performance in strategies during the outbreak of the pandemic. Therefore, we incorporate a trend component into the VMD-ACI-iLSTM model and examined its performance.

We construct an interval-based dummy variable based on the price movement of crude oil. If the opening price has continuously increased for three consecutive trading days, the left endpoint is set to 1. If it has continuously decreased for three consecutive trading days, the left endpoint is set to -1 . For all other cases where the opening price does not exhibit a clear trend, the left endpoint is set to 0. We use the same approach to construct the right endpoint based on the closing price.

Table 10

Comparison of the VMD-ACI-iLSTM model with and without trend data.

	U^I	ARV^I	MSE^I	MDE^I	$MAPE^I$	DA	AR
Original Model	0.8421	0.0945	0.9615	0.6388	0.01457	0.5167	61.19%
New Model	0.8413	0.0944	0.9583	0.6389	0.01454	0.5193	61.61%

“Original Model” represents the VMD-ACI-iLSTM model, while “New Model” represents the VMD-ACI-iLSTM model with trend data incorporated. “AR” is an abbreviation for “Annualized Return”.

This newly created interval variable allows us to capture the short-term trends in crude oil price movements and thus represents the market trend. Given that the iLSTM model is already effective in capturing short-term trends, we incorporate this market trend data into the ACI model to predict the low-frequency component. The performance is shown in Table 10.

We find that most error metrics experience a marginal improvement, indicating that incorporating the market trend data potentially enhances prediction accuracy. Nevertheless, the DM test results suggest that there is no significant difference between the predictions of the VMD-ACI-iLSTM model and the VMD-ACIX-iLSTM model (with the market trend data) at a 0.05 significance level. Thus, the inclusion of the market trend variable does not statistically enhance the prediction performance relative to the original hybrid model.

In summary, the incorporation of sentiment and trend data into our proposed VMD-ACI-iLSTM model does not appear imperative for price forecasting. Nonetheless, these data do exhibit a positive influence on DA and annualized return. Therefore, further exploration of effective sentiment and trend data proxies can be considered in the future.

7. Conclusion

In this paper, we propose a novel interval-based hybrid model for the prediction of crude oil prices. The proposed VMD-ACI-iLSTM model consists of three steps: First, Variational Mode Decomposition (VMD) segregates high and low-frequency components from the intrinsic series. Second, the ACI model forecasts the interval-valued time series of low-frequency components, while the iLSTM network predicts the interval-valued time series comprising high-frequency components. Last, a confluence of the two aforementioned components furnishes the final interval-valued prediction. Additionally, We develop a novel interval-based trading strategy for crude oil prices. We find that the proposed VMD-ACI-iLSTM model outperforms all benchmark models in terms of prediction and makes much more profits while taking less risks in terms of trading.

Three principal reasons underpin this phenomenon: firstly, interval-valued crude oil prices encapsulate more information compared to point-valued data within the equivalent period. Secondly, the VMD approach adeptly separates high- and low-frequency components from the indigenous series, enhancing modeling of discrete components and bolstering forecasting accuracy. Lastly, treating interval observations as an inseparable random set enables our approach to utilize the voluminous information within the interval data more efficaciously than other interval-valued models.

Whereas this study is circumscribed to univariate forecasting, crude oil prices are susceptible to numerous factors, including global economic growth and alterations in crude oil inventory. Prediction accuracy may be improved if such factors can be included in the proposed VMD-ACI-iLSTM model. Furthermore, a comparison of out-of-sample forecasts between the VMD-ACI-iLSTM model and other competing models in extant literature (Cai et al., 2000, 2009; Shu-ping et al., 2014; Yu and Ma, 2021) is warranted. Moreover, the framework proposed in this paper holds potential applicability in diverse markets, such as foreign exchange, gold, and stock markets. All these topics are questions open for future work.

CRedit authorship contribution statement

Li Zheng: Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Writing – original draft. **Yuying Sun:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Shouyang Wang:** Conceptualization, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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